

WASP-12b

R-0438

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_221203 · created 2026-07-19T13:32:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459915.87015 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1542 +/- 0.0092
Transit depth (Rp/Rs)^2	2.38 +/- 0.28 [%]
Orbital Inclination (inc)	78.92 +/- 2.35
Airmass coefficient 1 (a1)	1.314 +/- 0.011
Airmass coefficient 2 (a2)	-0.259 +/- 0.044
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	#3 - [453, 242]
Optimal Aperture	4.73
Optimal Annulus	8.34
Transit Duration (day)	0.1075 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1542 $\pm$ 0.0092 vs 0.1178 (deviation 3.93 σ)
Duration fit vs published	0.1075 d vs 0.125 d (deviation 3.55 σ)
Mid-time O-C	-1569.9 min
Depth SNR	16.7
Residual scatter	0.85 %

Light curve

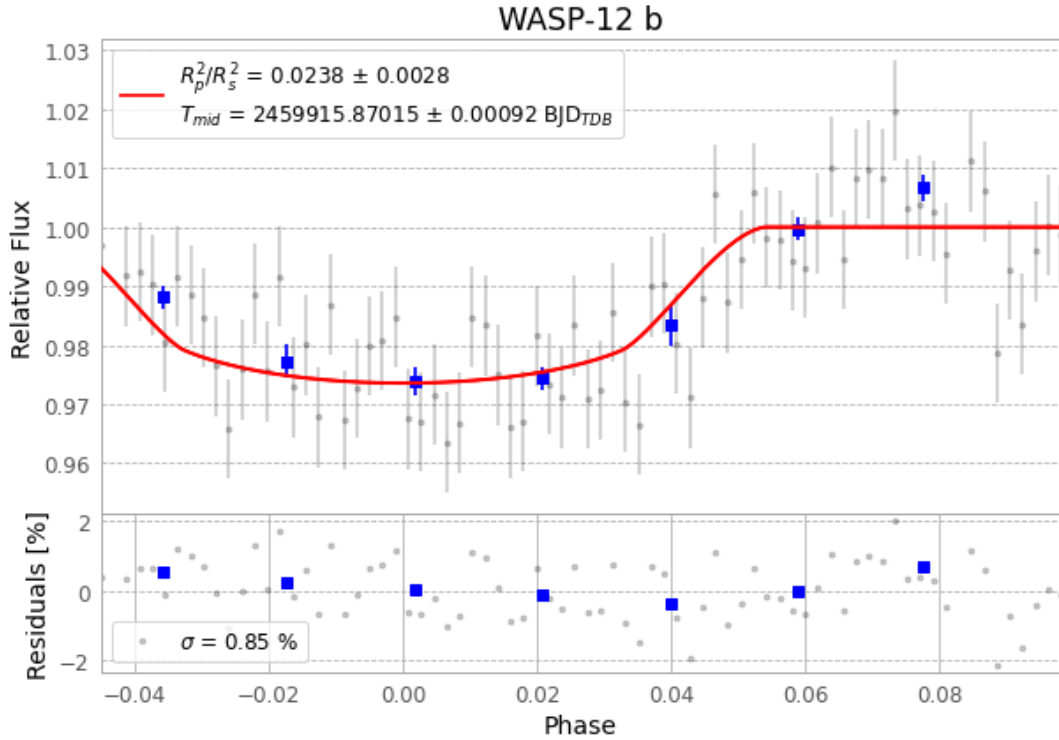


Figure 1 — FinalLightCurve_WASP-12 b_2022-12-02.png (SHA-256 f6f249e93ae7c4d4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 163], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b8f516dd650a2806...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

77 FITS frames (51 MB), WASP-12221202073016.FITS → WASP-12221202111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-221203.log (ddc670eb9696...), FinalLightCurve_WASP-12 b_2022-12-02.png (f6f249e93ae7...), FinalLightCurve_WASP-12 b_2022-12-02.csv (b4f40a678f9e...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 13:33Z.